

GYAN SETU

Chapter-4, Linear Equations in **Two Variables**

1. Introduction

In previous classes, you have studied linear equations in one variable such as $x + 1 = 0$ or $2y + 3 = 0$. A linear equation in one variable has only one solution, which can be represented on a number line. In this chapter, we extend this concept to linear equations in two variables, where we deal with equations that involve two unknown quantities. These equations are represented geometrically as straight lines on a graph.

A simple example is: $x + y = 10$. This means that the sum of two numbers is 10. If $x = 4$, then $y = 6$; if $x = 2$, then $y = 8$; and so on. Hence, there are infinitely many pairs (x, y) that satisfy the equation.

2. Linear Equations in Two Variables

An equation that can be expressed in the form $ax + by + c = 0$, where a , b , and c are real numbers and a and b are not both zero, is called a linear equation in two variables.

Examples of such equations:

- $2x + 3y = 4.37$
- $x - 4 = 3y$
- $4 = 5x - 3y$
- $2x = y$

Each of these equations can be rearranged into the standard form $ax + by + c = 0$:

$$-2x + 3y - 4.37 = 0 \quad (a = 2, b = 3, c = -4.37)$$

$$-x - 3y - 4 = 0 \quad (a = 1, b = -3, c = -4)$$

$$-5x - 3y - 4 = 0 \quad (a = 5, b = -3, c = -4)$$

$$-2x - y + 0 = 0 \quad (a = 2, b = -1, c = 0)$$

Example: Representing Real-life Situations

Example: In a One-day International Cricket match, two batsmen together scored 176 runs. If one batsman scored x runs and the other scored y runs, we can express this information as:

$$x + y = 176$$

This is a linear equation in two variables, where x and y represent the runs scored by each batsman respectively.

3. Solution of a Linear Equation in Two Variables

The solution of a linear equation in two variables is a pair of values (x, y) that satisfies the equation. When these values are substituted into the equation, both sides become equal.

Example: For the equation $2x + 3y = 12$, substituting $x = 3$ and $y = 2$ gives:

$$2 \times 3 + 3 \times 2 = 6 + 6 = 12$$

Hence, $(3, 2)$ is a solution.

Similarly, $(0, 4)$ and $(6, 0)$ are also solutions.

Therefore, a linear equation in two variables has infinitely many solutions, each represented by a point on the line.

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Coordinates of a Point

A point in the plane is represented as an ordered pair (x, y) :

To find the solutions of a linear equation in two variables, follow these steps:

1. Choose a value for one variable (say x).
2. Substitute this value in the equation.
3. Solve for the other variable (y).
4. Repeat this process for different values of x to get more solutions.

Example: Solve $x + 2y = 6$

If $x = 0$, then $y = 3 \rightarrow (0, 3)$

If $y = 0$, then $x = 6 \rightarrow (6, 0)$

If $y = 1$, then $x = 4 \rightarrow (4, 1)$

If $x = 2$, then $y = 2 \rightarrow (2, 2)$

Hence, four solutions are: $(0,3)$, $(6,0)$, $(4,1)$, $(2,2)$.

Tabular Representation of Solutions

x	y
0	3
6	0
4	1
2	2

4. Important Points to Remember

The solution of a linear equation in two variables is always a pair of numbers (x, y).

2. A linear equation in two variables represents a straight line when plotted on a graph.

3. Each point on the line is a solution of the equation.

4. If $x = 0$, the equation gives the y-intercept; if $y = 0$, the equation gives the x-intercept.

5. Such equations have infinitely many solutions because a line extends infinitely in both directions.

5. Applications in Real Life

Linear equations in two variables are widely used in everyday life. Here are some examples:

1. **Business:** Calculating cost and profit relationships.

Example: Total cost = Fixed cost + (Cost per unit $\times$ Number of units).

2. **Physics:** Relating quantities like distance, speed, and time (Distance = Speed $\times$ Time).

3. **Finance:** Calculating interest, budget planning, and savings relations.

4. **Daily Life:** Mixing two solutions, calculating expenses, etc.
